

ML Experiment Tracker

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Description

Lightweight experiment tracking system. Log hyperparameters, metrics, artifacts. Compare experiments with Flask dashboard. SQLite-backed, zero-config.

Tech Stack

flask, pandas, matplotlib, click, pydantic, pytest

Features

- Log parameters, metrics, artifacts
- Tag-based organization
- Flask dashboard with charts
- Metric visualization
- CLI for listing/exporting
- CSV/JSON export

Sample Input

Training run data

```
python cli.py experiments && python cli.py dashboard
```

Sample Output

Experiment comparison

```
Best: accuracy=0.94 (run_3, lr=0.001, epochs=50)
```

Project Files

```
.gitignore
cli.py
generate_synthetic_data.py
project-idea.json
pyproject.toml
README.md
requirements.txt
.github/workflows/test.yml
dashboard/app.py
dashboard/__init__.py
```

dashboard/templates/base.html
dashboard/templates/compare.html
dashboard/templates/compare_select.html
dashboard/templates/error.html
dashboard/templates/experiment.html
dashboard/templates/index.html
dashboard/templates/run.html
tests/conftest.py
tests/test_dashboard.py
tests/test_models.py
tests/test_store.py
tests/test_tracker.py
tests/test_visualizations.py
tests/__init__.py
tracker/models.py
tracker/store.py
tracker/tracker.py
tracker/visualizations.py
tracker/__init__.py

Test Results

PASSED

Run: 39 | Passed: 39 | Failed: 0

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